2-Neighbor Bootstrap Percolation on Odd Graphs

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Abstract

The r-neighbor bootstrap percolation process on a graph is a vertex-activation process that begins with a set of initially active vertices. In each subsequent round, every inactive vertex having at least r active neighbors becomes active. A set of initially active vertices whose activation eventually spreads to all vertices of the graph is called a percolating set. Let $m(G, r)$ denote the minimum size of a percolating set in the r-neighbor bootstrap percolation process on a graph G. In this paper, among other results, we prove that

$$\frac{k^2}{4} + \Omega(k) \leqslant m(\mathbb{O}_k, 2) \leqslant \frac{k^2}{3} + O(k),$$

where $\mathbb{O}_k$ denotes the odd graph on a ground set of cardinality $2k+1$. As a consequence, we confirm a conjecture posed in 2021 by Grippo, Pastine, Torres, Valencia-Pabon, and Vera.

Keywords: bipartite odd graph, r-neighbor bootstrap percolation, odd graph, percolating set.

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1. Introduction

Bootstrap percolation processes on graphs constitute a class of cellular automata, a concept introduced by von Neumann in 1966 [12]. One of the most extensively studied examples is the r-neighbor bootstrap percolation process, which was introduced by Chalupa, Leath, and Reich [6] in 1979 in the context of ferromagnetism. This process is closely connected to several areas of graph theory and discrete mathematics, including weak saturation and P_3-convexity. It has also appeared in the literature under various names, such as irreversible threshold process, influence propagation, and dynamic monopoly.

Throughout this paper, all graphs are finite, simple, and undirected. For a graph G, let $V(G)$ and $E(G)$ denote its vertex set and edge set, respectively. Given a nonnegative integer r, the *r-neighbor bootstrap percolation process* on a graph G begins with a set $V_0 \subseteq V(G)$ of initially active vertices. For each integer $i \geqslant 1$, the set of active vertices at round i is defined by

$$V_i = V_{i-1} \cup \Big\{ v \in V(G) \,\Big|\, v \text{ is adjacent to at least } r \text{ vertices in } V_{i-1} \Big\}.$$

A set V_0 is called a *percolating set* if $V_t = V(G)$ for some integer $t \geqslant 0$. We denote by $m(G, r)$ the minimum cardinality of a percolating set in the r-neighbor bootstrap percolation process on G. Percolating sets are sometimes referred to as *contagious sets*. Moreover, in the context of graph convexity, the parameter $m(G, 2)$ is known as the *P_3-hull number* of G.

An edge analogue of the r-neighbor bootstrap percolation process is the *r-edge bootstrap percolation process*. Given a nonnegative integer r and a graph G, the process begins with a set $E_0 \subseteq E(G)$ of initially active edges. For each integer $i \geqslant 1$, the set of active edges at round i is defined by

$$E_i = E_{i-1} \cup \Big\{ e \in E(G) \,\Big|\, \text{at least one endpoint of } e \text{ is incident to at least } r \text{ edges in } E_{i-1} \Big\}.$$

A set E_0 is called a *percolating set* if $E_t = E(G)$ for some integer $t \geqslant 0$. Let $m_e(G, r)$ denote the minimum cardinality of a percolating set in the r-edge bootstrap percolation process on G. The case $r = 2$ was studied by Lenormand and Zarcone [10] in 1984 under a different name.

Using a result from [8], we have

$$\frac{m_e(G,r)}{r} \leqslant m(G,r) \leqslant m_e(G,r) + \Big| \Big\{ v \in V(G) \,\Big|\, \deg(v) < r \Big\} \Big|, \tag{1}$$

where $\deg(v) = |\{x \in V(G) \,|\, x \text{ is adjacent to } v\}|$. A central problem in bootstrap percolation is to determine the extremal parameters $m(G, r)$ and $m_e(G, r)$. These parameters have been extensively studied across various graph families, including grids, tori, and Hamming graphs, see for example [1, 11, 13]. Additionally, this problem has been widely examined for random graphs, see for example [9].

By a result of Hambardzumyan, Hatami, and Qian [8], we have

$$\frac{m_e(G,r)}{r} \leqslant m(G,r) \leqslant m_e(G,r) + \Big| \Big\{ v \in V(G) \,\Big|\, \deg(v) < r \Big\} \Big|, \tag{2}$$

where

$$\deg(v) = \Big| \Big\{ x \in V(G) \mid x \text{ is adjacent to } v \Big\} \Big|.$$

Determining the extremal parameters $m(G, r)$ and $m_e(G, r)$ is one of the fundamental problems in bootstrap percolation theory. These parameters have been studied extensively for numerous graph classes, including grids, tori, and Hamming graphs; see, for example, [1, 11, 13]. They have also been investigated in various models of random graphs; see [9].

We next recall some standard notation and terminology. For two adjacent vertices u and v, we write $u \sim v$. Let k be a positive integer and set $M = \{1, \ldots, 2k+1\}$. The *odd graph* $\mathbb{O}_k$ is the graph whose vertices are the k-subsets of M, where two vertices A and B are adjacent if and only if $A \cap B = \varnothing$. The term "odd graph" is due to Biggs and Gardiner [5], who observe that each edge can be labeled by the unique element of M not contained in either endpoint, representing the "odd man out". Odd graphs form a highly symmetric subclass of Kneser graphs and exhibit remarkable algebraic and geometric properties not shared by general members of the Kneser family. The *bipartite odd graph* $\mathbb{B}_k$ is the graph whose vertex set consists of all k-subsets and all $(k+1)$-subsets of M, where two vertices A and B are adjacent if and only if $A \subsetneq B$. Equivalently, $\mathbb{B}_k$ is the bipartite double cover of $\mathbb{O}_k$. Recall that the bipartite double cover of a graph G with vertex set $\{v_1, \ldots, v_n\}$ is the bipartite graph with vertex set $\{v_1', \ldots, v_n'\} \cup \{v_1'', \ldots, v_n''\}$ in which v_i' is adjacent to v_j'' if and only if $v_i \sim v_j$ in G. Equivalently, the bipartite double cover of G is isomorphic to the direct product $G \times K_2$, where K_2 is the complete graph on two vertices. It is well known that both odd graphs and bipartite odd graphs are distance-regular graphs [3].

In this paper, we study the parameters $m(\mathbb{O}_k, 2)$, $m(\mathbb{B}_k, 2)$, and $m_e(\mathbb{B}_k, 2)$. We employ the polynomial method introduced by Hambardzumyan, Hatami, and Qian [8] to determine the exact values of $m(\mathbb{B}_k, 2)$ and $m_e(\mathbb{B}_k, 2)$, as established in the following theorem.

Theorem 1.1. *For every positive integer k,*

$$m_e(\mathbb{B}_k, 2) = k^2 + 2k + 3 \quad \textit{and} \quad m(\mathbb{B}_k, 2) = \left\lceil \frac{k^2 + 2k + 3}{2} \right\rceil.$$

Moreover, Grippo, Pastine, Torres, Valencia-Pabon, and Vera [7] conjectured that $m(\mathbb{O}_k, 2) = \Theta(k^2)$. We resolve this conjecture affirmatively by proving the following theorem.

Theorem 1.2. *For every positive integer $k \geqslant 10$,*

$$\frac{k^2 + 2k + 3}{4} \leqslant m(\mathbb{O}_k, 2) \leqslant \frac{k^2 + 5k + 3}{3}.$$

The paper is organized as follows. In Section 2, we present the necessary preliminaries and introduce equivalent definitions of odd graphs and bipartite odd graphs. In Section 3, we determine the exact values of $m(\mathbb{B}_k, 2)$ and $m_e(\mathbb{B}_k, 2)$ for all k. Finally, in Section 4, we establish upper and lower bounds on $m(\mathbb{O}_k, 2)$, thereby resolving a conjecture posed in [7].

2. Preliminaries

In this section, we provide the necessary preliminaries and background for the remainder of the article. We begin by presenting equivalent definitions of odd graphs and bipartite odd graphs that will be essential for our subsequent analysis.

Recall that k is a positive integer and let $M = \{1, \ldots, 2k+1\}$. Fix the bipartition $X = \{1, \ldots, k\}$ and $Y = \{k+1, \ldots, 2k+1\}$ of M. For every subset $A \subseteq M$, we write $A_X = A \cap X$ and $A_Y = A \cap Y$. Define

$$\mathscr{F} = \left\{ A \subseteq M \,\middle|\, |A_X| = |A_Y| \text{ or } |A_Y| = |A_X| + 1 \right\}.$$

We show that $\mathbb{B}_k$ is isomorphic to the graph with vertex set $\mathscr{F}$ in which two vertices A and B are adjacent if $|A \triangle B| = 1$, where $\triangle$ denotes the symmetric difference. To establish this isomorphism, consider the map $\varphi : \mathscr{F} \to V(\mathbb{B}_k)$ defined by

$$\varphi(A) = \begin{cases} (X \setminus A_X) \cup A_Y, & \text{if } |A_X| = |A_Y|, \\ A_X \cup (Y \setminus A_Y), & \text{if } |A_Y| = |A_X| + 1. \end{cases}$$

Thus, we obtain an equivalent description of $\mathbb{B}_k$. It follows immediately that $\mathbb{B}_k$ is an induced subgraph of the $(2k+1)$-dimensional hypercube. For any two adjacent vertices $A, B \in \mathbb{B}_k$, we assign to the edge $A \sim B$ the unique element of $A \triangle B$. This yields a proper edge-coloring of $\mathbb{B}_k$.

Let $\mathscr{F}'$ denote the subset of $\mathscr{F}$ consisting of all elements of size at most k. It is straightforward to verify that the restriction of φ to $\mathscr{F}'$ induces an isomorphism between $\mathbb{O}_k$ and the graph with vertex set $\mathscr{F}'$, where two vertices A and B are adjacent if $|A \triangle B| \in \{1, 2k\}$. This provides an equivalent definition of $\mathbb{O}_k$. Moreover, for any $A, B \in \mathscr{F}'$, the case $|A \triangle B| = 2k$ occurs if and only if $|A| = |B| = k$ and $A \cap B = \varnothing$.

We now recall a polynomial method introduced by Hambardzumyan, Hatami, and Qian [8]. We shall use this method to obtain a lower bound on $m_e(\mathbb{B}_k, 2)$ in Section 3. It is worth noting that Miralaei, Mohammadian, and Tayfeh-Rezaie [11] showed that this method is a special case of a general linear-algebraic lemma due to Balogh, Bollobás, Morris, and Riordan [2], which has been used to establish several results on weak saturation. We remark that the following formulation differs slightly from the original version in [8]; the present refined version appears in [4, 11].

Definition 2.1 ([8])**.** Let r be a nonnegative integer and let G be a graph equipped with a proper edge-coloring $c : E(G) \to \mathbb{R}$. Let $W_c(G, r)$ be the vector space over $\mathbb{R}$ consisting of all functions $\varphi : E(G) \to \mathbb{R}$ for which there exist polynomials $\{P_v(x)\}_{v \in V(G)}$ satisfying

(i) $\deg P_v(x) \leqslant r - 1$ for every vertex $v \in V(G)$;

(ii) $P_u(c(uv)) = P_v(c(uv)) = \varphi(uv)$ for each edge $uv \in E(G)$.

It is said that the polynomials $\big\{P_v(x)\big\}_{v \in V(G)}$ *recognize* φ. Notice that we adopt the convention that the degree of the zero polynomial is -1.

The next theorem provides a linear-algebraic lower bound on $m_e(G, r)$ for any graph G and any nonnegative integer r. This result plays a central role in our arguments.

Theorem 2.2 ([8])**.** *Let r be a nonnegative integer and let $c : E(G) \to \mathbb{R}$ be a proper edge-coloring of a graph G. Then, $m_e(G, r) \geqslant \dim W_c(G, r)$.*

The next proposition is folklore, but we include a short proof for completeness.

Proposition 2.3. *Let G be a graph and let H be the bipartite double cover of G. Then, $m(H, r) \leqslant 2m(G, r)$ for every positive integer r.*

Proof. Let $V(G) = \{v_1, \ldots, v_n\}$ and let $V(H) = \{v'_1, \ldots, v'_n\} \cup \{v''_1, \ldots, v''_n\}$, where v'_i and v''_i are the vertices of H corresponding to $v_i \in V(G)$ for each i. For any subset $V \subseteq V(G)$, we define

$V' = \{v' \mid v \in V\}$ and $V'' = \{v'' \mid v \in V\}$. Assume that V_0 is a percolating set in the r-neighbor bootstrap percolation process on G, and let V_t denote the set of vertices activated after t rounds, where $t \geqslant 1$. We claim that $V_0' \cup V_0''$ is a percolating set in the r-neighbor bootstrap percolation process on H. Indeed, at each round $t \geqslant 0$, the vertices in V_t' activate all vertices in V_{t+1}'', while the vertices in V_t'' activate all vertices in V_{t+1}'. Since V_0 is a percolating set of G, this process eventually activates all vertices of H. The result follows immediately. □

3. Bipartite odd graphs

In this section, we determine the exact values of the vertex and edge percolation numbers of the bipartite odd graph $\mathbb{B}_k$. We begin by establishing a lower bound on the edge percolation number using the polynomial method described in Section 2. The matching upper bound is then obtained by an explicit construction of a percolating edge set. Combining these two bounds yields the exact value of $m_e(\mathbb{B}_k, 2)$. Finally, we use the general relation (2) between vertex and edge percolation numbers to derive the corresponding value of $m(\mathbb{B}_k, 2)$.

In the following lemma, c denotes the proper edge-coloring of $\mathbb{B}_k$ defined in Section 2.

Lemma 3.1. *For every positive integer k,*

$$\dim W_c(\mathbb{B}_k, 2) \geqslant k^2 + 2k + 3.$$

Proof. The proof consists of three steps. First, we construct a collection of polynomial assignments to the vertices of $\mathbb{B}_k$. Next, we verify that these assignments satisfy Definition 2.1. Finally, we show that the corresponding vectors are linearly independent, yielding the desired lower bound.

We construct three families of polynomial assignments, each consisting of polynomials of degree at most one. The total number of assignments is

$$2 + (k+1) + (k^2 + k) = k^2 + 2k + 3.$$

Family 1 (2 assignments). Assign the constant polynomial 1 and the linear polynomial x to every vertex.

Family 2 ($k+1$ assignments). For each $b \in Y$, define

$$P_A^b(x) = \begin{cases} x - b, & b \in A, \\ 0, & b \notin A, \end{cases} \qquad A \in V(\mathbb{B}_k).$$

Family 3 ($k^2 + k$ assignments). For $a \in X$ and $b \in Y$, define

$$Q_A^{ab}(x) = \begin{cases} h_A^{ab}(x - a), & |A_Y| = |A_X| \text{ and } a, b \in A, \\ h_A^{ab}(x - b), & |A_Y| = |A_X| + 1 \text{ and } a, b \in A, \\ 0, & \text{otherwise,} \end{cases}$$

where

$$h_A^{ab} = \prod_{u \in A_X \setminus \{a\}} \frac{u-b}{u-a} \prod_{v \in A_Y \setminus \{b\}} \frac{v-a}{v-b}.$$

We now verify that these assignments satisfy Definition 2.1. Condition (i) is immediate for all three families. It remains to check condition (ii).

Family 1. For every edge AB, the same polynomial is assigned to both endpoints. Hence condition (ii) holds trivially.

Family 2. Fix $b \in Y$, and let AB be an edge. Without loss of generality, assume $B = A \cup \{p\}$ for some $p \in M \setminus A$.

If $p = b$, then $P_A^b(x) = 0$ and $P_B^b(x) = x - b$, so

$$P_A^b(b) = P_B^b(b) = 0.$$

If $p \neq b$, then either $b \in A \cap B$ or $b \notin A \cup B$. In both cases the same polynomial is assigned to A and B, and condition (ii) follows immediately.

Family 3. Fix $a \in X$ and $b \in Y$, and let AB be an edge with $B = A \cup \{p\}$.

If $\{a, b\} \not\subseteq B$, then $\{a, b\} \not\subseteq A$ as well, and therefore

$$Q_A^{ab}(x) = Q_B^{ab}(x) = 0.$$

Hence condition (ii) is satisfied. We may therefore assume that $a, b \in B$.

Case 1: $p = a$.

Since $a \notin A$, we have $Q_A^{ab}(x) = 0$. Moreover, $|B_Y| = |B_X|$, so

$$Q_B^{ab}(x) = h_B^{ab}(x - a).$$

Consequently,

$$Q_A^{ab}(a) = Q_B^{ab}(a) = 0.$$

Case 2: $p = b$.

Since $b \notin A$, we have $Q_A^{ab}(x) = 0$. Furthermore, $|B_Y| = |B_X| + 1$, so

$$Q_B^{ab}(x) = h_B^{ab}(x - b).$$

Hence

$$Q_A^{ab}(b) = Q_B^{ab}(b) = 0.$$

Case 3: $p \notin \{a, b\}$ and $p \in X$.

In this case $|A_Y| = |A_X| + 1$, and therefore

$$Q_A^{ab}(x) = h_A^{ab}(x - b), \qquad Q_B^{ab}(x) = h_B^{ab}(x - a).$$

A direct computation gives

$$h_B^{ab} = h_A^{ab} \frac{p-b}{p-a}.$$

It follows that
$$Q_A^{ab}(p) = Q_B^{ab}(p).$$

Case 4: $p \notin \{a,b\}$ and $p \in Y$.

Now $|A_Y| = |A_X|$, so
$$Q_A^{ab}(x) = h_A^{ab}(x-a), \qquad Q_B^{ab}(x) = h_B^{ab}(x-b).$$
Moreover,
$$h_B^{ab} = h_A^{ab}\frac{p-a}{p-b},$$
which again yields
$$Q_A^{ab}(p) = Q_B^{ab}(p).$$

Thus condition (ii) holds in all cases.

It remains to determine the dimension of the resulting vector space. By Definition 2.1, each polynomial assignment gives rise to a vector indexed by the edges of $\mathbb{B}_k$, whose entry on an edge e is obtained by evaluating the corresponding endpoint polynomials at the color $c(e)$.

Let $\mathbf{V}_1, \mathbf{V}_2$ denote the vectors arising from Family 1, let $\mathbf{W}_b$ ($b \in Y$) denote the vectors arising from Family 2, and let $\mathbf{U}_{ab}$ ($a \in X$, $b \in Y$) denote the vectors arising from Family 3.

Suppose that
$$\sum_{i=1}^{2} \alpha_i \mathbf{V}_i + \sum_{b\in Y} \beta_b \mathbf{W}_b + \sum_{a\in X,\, b\in Y} \lambda_{ab}\mathbf{U}_{ab} = 0. \tag{3}$$

Consider the following sets of edges:

1. E_0: the set of all edges incident with $\varnothing$;
2. E_1: the set of all edges joining $\{b\}$ and $\{a,b\}$, where $a \in X$ and $b \in Y$;
3. E_2: the set of all edges joining $\{a,b\}$ and $\{a,b,b'\}$, where $a \in X$ and $b, b' \in Y$.

For every $e \in E_0$,
$$\mathbf{V}_1(e) = 1, \qquad \mathbf{V}_2(e) = c(e),$$
while
$$\mathbf{W}_b(e) = 0, \qquad \mathbf{U}_{ab}(e) = 0.$$
Since $|E_0| > 1$, equation (3) implies
$$\alpha_1 = \alpha_2 = 0.$$

Next, let
$$e = \big\{\{b_0\}, \{a_0, b_0\}\big\} \in E_1.$$
Then $\mathbf{U}_{ab}(e) = 0$ for all a, b, and
$$\mathbf{W}_b(e) = \begin{cases} a_0 - b_0, & b = b_0,\\ 0, & b \neq b_0.\end{cases}$$

Hence equation (3) yields

$$\beta_{b_0} = 0$$

for every $b_0 \in Y$.

Finally, let

$$e = \big\{\{a_0, b_0\}, \{a_0, b_0, b_1\}\big\} \in E_2.$$

Then

$$\mathbf{U}_{ab}(e) = \begin{cases} b_1 - a_0, & a = a_0 \text{ and } b = b_0, \\ 0, & \text{otherwise.} \end{cases}$$

Therefore,

$$\lambda_{a_0 b_0} = 0$$

for every $a_0 \in X$ and $b_0 \in Y$.

We conclude that all coefficients in (3) vanish. Hence the $k^2 + 2k + 3$ vectors obtained from the three families are linearly independent. Therefore,

$$\dim W_c(\mathbb{B}_k, 2) \geqslant k^2 + 2k + 3,$$

as required. □

Lemma 3.2. *For every positive integer k, $m_e(\mathbb{B}_k, 2) \leqslant k^2 + 2k + 3$.*

Proof. Let A be the empty set vertex of $\mathbb{B}_k$. For each $i \geqslant 0$, let $\mathcal{V}_i$ denote the set of vertices at distance i from A. It is easily seen that every vertex in $\mathcal{V}_i$ has at least two neighbors in $\mathcal{V}_{i-1}$ whenever $i \geqslant 3$. Consequently, if all edges incident with vertices in $\mathcal{V}_0 \cup \mathcal{V}_1 \cup \mathcal{V}_2$ are activated, then every remaining edge will eventually become activated as well.

To achieve this, we initially activate

- two arbitrary edges incident with A,
- for each vertex $B \in \mathcal{V}_1$, one arbitrary edge joining B to a vertex in $\mathcal{V}_2$,
- for each vertex $C \in \mathcal{V}_2$, one arbitrary edge joining C to a vertex in $\mathcal{V}_3$.

This gives an initial set of $k^2 + 2k + 3$ activated edges. It is straightforward to verify that this initialization activates all edges incident with vertices in $\mathcal{V}_0 \cup \mathcal{V}_1 \cup \mathcal{V}_2$. □

The following theorem follows directly from Lemmas 3.1 and 3.2 and Theorem 2.2.

Theorem 3.3. *For every positive integer k, $m_e(\mathbb{B}_k, 2) = k^2 + 2k + 3$.*

Theorem 3.4. *For every positive integer k,*

$$m(\mathbb{B}_k, 2) = \left\lceil \frac{k^2 + 2k + 3}{2} \right\rceil.$$

Proof. By (2) and Theorem 3.3,

$$m(\mathbb{B}_k, 2) \geqslant \left\lceil \frac{k^2+2k+3}{2} \right\rceil.$$

Therefore, it suffices to construct a percolating set of cardinality

$$\left\lceil \frac{k^2+2k+3}{2} \right\rceil.$$

We distinguish according to the parity of k.

Case 1: k is odd.

Define

$$\begin{aligned}
\mathcal{P}_1 &= \big\{\{k+1\},\{k+2\}\big\},\\
\mathcal{P}_2 &= \big\{\{1,i\} : k+3 \leqslant i \leqslant 2k+1\big\},\\
\mathcal{P}_3 &= \big\{\{i,k+1,k+2\} : 1 \leqslant i \leqslant k\big\},\\
\mathcal{P}_4 &= \Big\{\{i,2j,2j+1\} : 2 \leqslant i \leqslant k,\ \frac{k+3}{2} \leqslant j \leqslant k\Big\}.
\end{aligned}$$

Let

$$\mathcal{P} = \mathcal{P}_1 \cup \mathcal{P}_2 \cup \mathcal{P}_3 \cup \mathcal{P}_4.$$

A straightforward calculation yields

$$|\mathcal{P}| = 2 + (k-1) + k + \frac{(k-1)^2}{2} = \frac{k^2+2k+3}{2}.$$

Thus it remains to show that $\mathcal{P}$ percolates.

For every $1 \leqslant i \leqslant k$,

$$\{i,k+1\} \sim \{k+1\} \qquad \text{and} \qquad \{i,k+1\} \sim \{i,k+1,k+2\}.$$

Since both neighbors belong to $\mathcal{P}$, every vertex $\{i,k+1\}$ becomes active. Similarly,

$$\{i,k+2\} \sim \{k+2\} \qquad \text{and} \qquad \{i,k+2\} \sim \{i,k+1,k+2\},$$

and therefore every vertex $\{i,k+2\}$ becomes active.

The empty vertex $\varnothing$ is adjacent to both $\{k+1\}$ and $\{k+2\}$. Since these vertices are active, $\varnothing$ acquires two active neighbors and becomes active.

Now fix $j \in Y \setminus \{k+1,k+2\}$. Since

$$\{j\} \sim \{1,j\} \qquad \text{and} \qquad \{j\} \sim \varnothing,$$

and both neighbors are active, the vertex $\{j\}$ becomes active. Consequently, every singleton vertex is active.

We next show that every vertex of size two becomes active. Let $\{a,b\}$ be a vertex with

$$2 \leqslant a \leqslant k, \qquad k+3 \leqslant b \leqslant 2k+1.$$

Then

$$\{a,b\} \sim \{b\} \qquad \text{and} \qquad \{a,b\} \sim \{a,b,s\},$$

where

$$s = \begin{cases} b+1, & \text{if } b \text{ is even}, \\ b-1, & \text{if } b \text{ is odd}. \end{cases}$$

The vertex $\{b\}$ is already active, while $\{a,b,s\} \in \mathcal{P}_4$. Hence $\{a,b\}$ has two active neighbors and becomes active.

To complete the proof, we proceed by induction on the size of a vertex. We have already shown that every vertex of size at most 2 is active. Assume that every vertex of size at most t, where $t \geqslant 2$, is active. Let A be a vertex of size $t+1$. Since A has at least two neighbors of size t, both active by the induction hypothesis, the vertex A becomes active. Therefore every vertex of $\mathbb{B}_k$ eventually becomes active, proving that $\mathcal{P}$ is a percolating set.

Case 2: k is even.

Define

$$\begin{aligned}
\mathcal{P}_1 &= \big\{\{k+1\},\{k+2\},\{k+3\}\big\},\\
\mathcal{P}_2 &= \big\{\{1,j\} : k+4 \leqslant j \leqslant 2k+1\big\},\\
\mathcal{P}_3 &= \big\{\{i,k+2,k+3\} : 1 \leqslant i \leqslant k\big\},\\
\mathcal{P}_4 &= \Big\{\{i,2j,2j+1\} : 2 \leqslant i \leqslant k,\ \frac{k+4}{2} \leqslant j \leqslant k\Big\},\\
\mathcal{P}_5 &= \Big\{\{2i-1,2i,k+1,k+2\} : 1 \leqslant i \leqslant \frac{k}{2}\Big\}.
\end{aligned}$$

Let

$$\mathcal{P} = \mathcal{P}_1 \cup \mathcal{P}_2 \cup \mathcal{P}_3 \cup \mathcal{P}_4 \cup \mathcal{P}_5.$$

Since

$$|\mathcal{P}_1| = 3, \qquad |\mathcal{P}_2| = k-2, \qquad |\mathcal{P}_3| = k,$$
$$|\mathcal{P}_4| = (k-1)\frac{k-2}{2}, \qquad |\mathcal{P}_5| = \frac{k}{2},$$

we obtain

$$|\mathcal{P}| = 3 + (k-2) + k + \frac{(k-1)(k-2)}{2} + \frac{k}{2} = \frac{k^2+2k+4}{2} = \left\lceil \frac{k^2+2k+3}{2} \right\rceil.$$

We show that $\mathcal{P}$ is a percolating set.

For every $1 \leqslant i \leqslant k$,

$$\{i,k+2\} \sim \{k+2\} \qquad \text{and} \qquad \{i,k+2\} \sim \{i,k+2,k+3\},$$

hence every vertex $\{i, k+2\}$ becomes active. Similarly,

$$\{i, k+3\} \sim \{k+3\} \qquad \text{and} \qquad \{i, k+3\} \sim \{i, k+2, k+3\},$$

and therefore every vertex $\{i, k+3\}$ becomes active.

Since both $\{k+2\}$ and $\{k+3\}$ are active, the vertex $\varnothing$ becomes active as well.

For every $j \geqslant k+4$,

$$\{j\} \sim \{1, j\} \qquad \text{and} \qquad \{j\} \sim \varnothing.$$

Since both neighbors are active, every such singleton vertex $\{j\}$ becomes active.

Next, for each $1 \leqslant i \leqslant k/2$,

$$\{2i, k+1, k+2\} \sim \{2i, k+2\} \qquad \text{and} \qquad \{2i, k+1, k+2\} \sim \{2i-1, 2i, k+1, k+2\},$$

so $\{2i, k+1, k+2\}$ becomes active. Likewise,

$$\{2i-1, k+1, k+2\} \sim \{2i-1, k+2\} \qquad \text{and} \qquad \{2i-1, k+1, k+2\} \sim \{2i-1, 2i, k+1, k+2\},$$

and hence $\{2i-1, k+1, k+2\}$ becomes active. Consequently, every vertex $\{i, k+1, k+2\}$, $1 \leqslant i \leqslant k$, becomes active.

Since

$$\{i, k+1\} \sim \{k+1\} \qquad \text{and} \qquad \{i, k+1\} \sim \{i, k+1, k+2\},$$

and both neighbors are active, every vertex $\{i, k+1\}$ becomes active.

We next show that all vertices of size two become active. Let $\{a, b\}$ be a vertex with

$$2 \leqslant a \leqslant k, \qquad k+4 \leqslant b \leqslant 2k+1.$$

Then

$$\{a, b\} \sim \{b\} \qquad \text{and} \qquad \{a, b\} \sim \{a, b, s\},$$

where

$$s = \begin{cases} b+1, & \text{if } b \text{ is even}, \\ b-1, & \text{if } b \text{ is odd}. \end{cases}$$

As before, $\{b\}$ is active and $\{a, b, s\} \in \mathcal{P}_4$, so $\{a, b\}$ becomes active.

Finally, the same inductive argument used in the odd case shows that every vertex of size at least three eventually becomes active. Hence $\mathcal{P}$ is a percolating set.

Combining both cases establishes the theorem. $\square$

4. Odd graphs

In this section we establish an upper bound on the minimum size of a percolating set in the odd graph $\mathbb{O}_k$. We begin with two auxiliary lemmas. The first shows how a percolating set consisting entirely of 2-sets can be transformed into a substantially smaller percolating set, while the second proves that the collection of all 2-sets already forms a percolating set. These results will be combined in the proof of our main theorem.

Lemma 4.1. *Every percolating set of size m consisting only of 2-sets in $\mathcal{O}_k$ gives rise to a percolating set of size at most $(m+3k+2)/2$.*

Proof. Let $\mathcal{P}$ be a percolating set consisting only of 2-sets. For each $i \in \{1, \dots, k\}$, define

$$Y_i = \{ j : \{i, j\} \in \mathcal{P} \}.$$

Partition Y_i arbitrarily into disjoint 2-sets

$$Y_i^1, Y_i^2, \dots$$

and, if $|Y_i|$ is odd, one additional 1-set Y_i^0.

For every part Y_i^ℓ, let

$$Z_i^\ell = \{i\} \cup Y_i^\ell.$$

Thus each Z_i^ℓ is either a 3-set or a 2-set.

Now define

$$\mathcal{P}' = \{ Z_i^\ell : 1 \leqslant i \leqslant k,\ \ell \geqslant 0 \} \cup \{\{j\} : k+1 \leqslant j \leqslant 2k+1\}.$$

We first estimate its size. For each i, the number of sets Z_i^ℓ is

$$\left\lceil \frac{|Y_i|}{2} \right\rceil \leqslant \frac{|Y_i|+1}{2}.$$

Since

$$\sum_{i=1}^{k} |Y_i| = m,$$

it follows that

$$|\mathcal{P}'| \leqslant \sum_{i=1}^{k} \frac{|Y_i|+1}{2} + (k+1) = \frac{m+k}{2} + k + 1 = \frac{m+3k+2}{2}.$$

It remains to show that $\mathcal{P}'$ percolates. Initially activate all sets in $\mathcal{P}'$. Consider an arbitrary set $\{i, j\} \in \mathcal{P}$. Since $j \in Y_i$, there exists ℓ such that $j \in Y_i^\ell$. Hence $\{i, j\} \subseteq Z_i^\ell$.

If $\{i, j\} = Z_i^\ell$ for some ℓ, then $\{i, j\} \in \mathcal{P}'$ and so it is initially activated. Otherwise, $\{i, j\} \subsetneqq Z_i^\ell$, for some 3-set Z_i^ℓ. Thus $\{i, j\}$ has two activated neighbors, namely Z_i^ℓ and $\{j\}$, and therefore becomes activated.

Consequently, every set in $\mathcal{P}$ is eventually activated. Since $\mathcal{P}$ is a percolating set, activation of all sets in $\mathcal{P}$ implies activation of the entire graph. Therefore $\mathcal{P}'$ is also a percolating set. $\square$

Lemma 4.2. *Let $k \geqslant 2$. The set of all vertices of size two in $\mathcal{O}_k$ is a percolating set.*

Proof. Initially, all vertices of size two are active. Let A be a vertex of size 1. Since A has exactly k neighbors of size 2, it acquires k active neighbors and hence becomes active. Consequently, the empty-set vertex $\varnothing$ has exactly $k+1$ active neighbors of size 1, and therefore it also becomes active.

We now consider vertices of size at least 3. Proceed by induction on the size of the vertex. Assume that every vertex of size at most t is active, where $t \geqslant 2$, and let A be a vertex of size $t+1$. Since A has at least two neighbors of size t, and all vertices of size t are active by the induction hypothesis, A has at least two active neighbors. Hence A becomes active.

Therefore every vertex of size $t+1$ is active. By induction, every vertex of $\mathbb{O}_k$ eventually becomes active. Thus, the set of all vertices of size two is a percolating set. □

Theorem 4.3. *For every positive integer $k \geqslant 10$,*

$$m(\mathbb{O}_k, 2) \leqslant \frac{k^2+5k+3}{3}.$$

Proof. Let

$$s = \left\lfloor \frac{k-1}{3} \right\rfloor \qquad \text{and} \qquad t = \left\lfloor \frac{k}{2} \right\rfloor - 1 - s.$$

Then

$$k = \begin{cases} 2s+2t+2, & \text{if } k \text{ is even,} \\ 2s+2t+3, & \text{if } k \text{ is odd.} \end{cases} \tag{4}$$

We partition X and Y as follows:

$$\begin{aligned} X_1 &= \{1, \dots, s\}, \\ X_2 &= \{s+1, \dots, 2s+1\}, \\ X_3 &= \{2s+2, \dots, k\}, \\ Y_1 &= \{k+1, \dots, k+s+1\}, \\ Y_2 &= \{k+s+2, \dots, 2k+1\}. \end{aligned}$$

Thus

$$|X_1| = s, \qquad |X_2| = s+1, \qquad |X_3| = k-2s-1,$$

and

$$|Y_1| = s+1, \qquad |Y_2| = k-s.$$

We initially activate all vertices in

$$(X_1 \times Y) \cup (X_2 \times Y_1) \cup (X_3 \times Y_2).$$

We shall prove that every vertex in

$$(X_2 \times Y_2) \cup (X_3 \times Y_1)$$

also becomes activated. Consequently, all vertices of size two become active, and Lemma 4.2 then implies that the entire graph percolates.

The number of initially activated vertices is

$$\begin{aligned} \big|(X_1 \times Y) \cup (X_2 \times Y_1) \cup (X_3 \times Y_2)\big| &= |X_1 \times Y| + |X_2 \times Y_1| + |X_3 \times Y_2| \\ &= s(k+1) + (s+1)^2 + (k-2s-1)(k-s) \\ &= k^2 - 2ks + 3s^2 + 4s - k + 1. \end{aligned}$$

Define

$$f(k,s) = k^2 - 2ks + 3s^2 + 4s - k + 1.$$

A straightforward calculation shows that

$$f\left(k, \frac{k-3}{3}\right) = f\left(k, \frac{k-1}{3}\right) = \frac{2k^2+k}{3},$$

while

$$f\left(k, \frac{k-2}{3}\right) = \frac{2k^2+k-1}{3}.$$

Since $s = \lfloor (k-1)/3 \rfloor$, it follows that

$$f(k,s) \leqslant \frac{2k^2+k}{3}.$$

Therefore, by Lemma 4.1, it suffices to prove that the above family of initially activated vertices indeed percolates.

We first show that every vertex A with $|A| \geqslant 2$ and

$$A_X \times A_Y \subseteq (X_1 \times Y) \cup (X_2 \times Y_1) \cup (X_3 \times Y_2)$$

becomes activated. We proceed by induction on $|A|$.

The claim is immediate when $|A| = 2$, since all such vertices are initially activated. Let $|A| \geqslant 3$, and assume that every vertex satisfying the above containment condition and having size $|A| - 1$ is activated.

If $|A_Y| = |A_X| + 1$, then $|A_Y| \geqslant 2$. Choose distinct elements $b_1, b_2 \in A_Y$. The vertices

$$A_X \cup (A_Y \setminus \{b_1\}) \qquad \text{and} \qquad A_X \cup (A_Y \setminus \{b_2\})$$

both satisfy the containment condition and have size $|A| - 1$. By the induction hypothesis, they are activated. Since they are neighbors of A, the vertex A acquires two activated neighbors and therefore becomes activated.

Similarly, if $|A_Y| = |A_X|$, then $|A_X| \geqslant 2$. Choose distinct elements $a_1, a_2 \in A_X$. The vertices

$$(A_X \setminus \{a_1\}) \cup A_Y \qquad \text{and} \qquad (A_X \setminus \{a_2\}) \cup A_Y$$

both satisfy the containment condition and have size $|A| - 1$. Hence they are activated by the induction hypothesis, and so A becomes activated as well.

For nonnegative integers x_1, x_2, x_3, y_1, y_2, let

$$(x_1, x_2, x_3, y_1, y_2)$$

denote the family of all vertices A satisfying

$$|A \cap X_i| = x_i \quad (i = 1,2,3) \qquad \text{and} \qquad |A \cap Y_j| = y_j \quad (j = 1,2).$$

We write

$$(x_1, x_2, x_3, y_1, y_2) \xrightarrow{\ i\ } (x_1', x_2', x_3', y_1', y_2')$$

to indicate that every vertex in $(x_1', x_2', x_3', y_1', y_2')$ has exactly i neighbors in $(x_1, x_2, x_3, y_1, y_2)$. Consequently, if all vertices in $(x_1, x_2, x_3, y_1, y_2)$ are activated and $i \geqslant 2$, then every vertex in $(x_1', x_2', x_3', y_1', y_2')$ becomes activated.

By the previous paragraph, every vertex in

$$(s-t, 0, 2t+1, 0, s+t+1)$$

is activated when k is even, while every vertex in

$$(s-t-1, 0, 2t+2, 0, s+t+2)$$

is activated when k is odd. In both cases, (4) implies that these vertices have size k.

When k is even,

$$(s-t, 0, 2t+1, 0, s+t+1) \xrightarrow{s+t+2} (t, s+1, 0, s+1, t),$$

whereas for odd k,

$$(s-t-1, 0, 2t+2, 0, s+t+2) \xrightarrow{s-t} (t, s+1, 0, s+1, t+1) \xrightarrow{s+t+3} (t, s+1, 0, s+1, t).$$

Hence every vertex in $(t, s+1, 0, s+1, t)$ becomes activated. The above argument requires $s-t \geqslant 2$ when k is odd and $s-t \geqslant 1$ in general (see below). Both conditions are satisfied for $k \geqslant 10$.

We next obtain the activation of $(1, s+1, 0, s+1, 1)$ through the chain

$$\begin{aligned}
&(t, s+1, 0, s+1, t) \xrightarrow{s-t+1} (t-1, s+1, 0, s+1, t) \xrightarrow{k-s-t+1} \\
&(t-1, s+1, 0, s+1, t-1) \xrightarrow{s-t+2} (t-2, s+1, 0, s+1, t-1) \xrightarrow{k-s-t+2} \\
&(t-2, s+1, 0, s+1, t-2) \xrightarrow{s-t+3} \cdots \xrightarrow{k-s-1} (1, s+1, 0, s+1, 1).
\end{aligned}$$

Hence all vertices in $(1, s+1, 0, s+1, 1)$ become activated.

Furthermore,

$$\begin{aligned}
&(1, s+1, 0, s+1, 1) \xrightarrow{1} (1, s, 0, s+1, 1), \\
&(1, s, 0, s+1, 0) \xrightarrow{1} (1, s, 0, s+1, 1).
\end{aligned}$$

Since all vertices in $(1, s+1, 0, s+1, 1)$ and $(1, s, 0, s+1, 0)$ are already active, it follows that every vertex in $(1, s, 0, s+1, 1)$ becomes activated.

We may now iterate the following procedure:

$$\begin{aligned}
&(1, s, 0, s+1, 1) \xrightarrow{2} (2, s, 0, s+1, 1) \xrightarrow{2} \\
&(2, s-1, 0, s+1, 1) \xrightarrow{3} (3, s-1, 0, s+1, 1) \xrightarrow{3} \\
&(3, s-2, 0, s+1, 1) \xrightarrow{4} \cdots \xrightarrow{s} (s, 1, 0, s+1, 1).
\end{aligned}$$

Consequently, all vertices in $(s, 1, 0, s+1, 1)$ are activated.

Next,

$$(s, 1, 0, s+1, 1) \xrightarrow{\quad 1 \quad} (s, 1, 0, s, 1),$$
$$(s, 0, 0, s, 1) \xrightarrow{\quad 1 \quad} (s, 1, 0, s, 1),$$

and therefore every vertex in $(s, 1, 0, s, 1)$ is activated. Similarly,

$$(s, 1, 0, s, 1) \xrightarrow{\quad 1 \quad} (s-1, 1, 0, s, 1),$$
$$(s-1, 1, 0, s, 0) \xrightarrow{\quad 1 \quad} (s-1, 1, 0, s, 1),$$

which implies that all vertices in $(s-1, 1, 0, s, 1)$ are activated.

Finally,

$$\begin{aligned}
&(s-1, 1, 0, s, 1) \xrightarrow{\quad 2 \quad} (s-1, 1, 0, s-1, 1) \xrightarrow{\quad 2 \quad} \\
&(s-2, 1, 0, s-1, 1) \xrightarrow{\quad 3 \quad} (s-2, 1, 0, s-2, 1) \xrightarrow{\quad 3 \quad} \\
&(s-3, 1, 0, s-2, 1) \xrightarrow{\quad 4 \quad} \cdots \xrightarrow{\quad s \quad} (1, 1, 0, 1, 1) \xrightarrow{\quad s \quad} (0, 1, 0, 1, 1) \xrightarrow{\quad s+1 \quad} (0, 1, 0, 0, 1).
\end{aligned}$$

Thus every vertex in $(0, 1, 0, 0, 1)$, equivalently every vertex in $X_2 \times Y_2$, becomes activated.

Combining this with the argument used earlier, we conclude that every vertex A with $|A| \geqslant 2$ and

$$A_X \times A_Y \subseteq ((X_1 \cup X_2) \times Y) \cup (X_3 \times Y_2)$$

is eventually activated.

It remains to activate the vertices in $(0, 0, 1, 1, 0)$, equivalently those in $X_3 \times Y_1$.

If k is even, then

$$(s-t, 1, 2t, 0, s+t+1) \xrightarrow{\quad s+t+2 \quad} (t, s, 1, s+1, t),$$

while for odd k,

$$(s-t-1, 1, 2t+1, 0, s+t+2) \xrightarrow{\quad s-t \quad} (t, s, 1, s+1, t+1) \xrightarrow{\quad s+t+3 \quad} (t, s, 1, s+1, t).$$

Hence every vertex in $(t, s, 1, s+1, t)$ becomes activated.

Proceeding as before, we obtain

$$\begin{aligned}
&(t, s, 1, s+1, t) \xrightarrow{\quad s-t+1 \quad} (t-1, s, 1, s+1, t) \xrightarrow{\quad k-s-t+1 \quad} \\
&(t-1, s, 1, s+1, t-1) \xrightarrow{\quad s-t+2 \quad} (t-2, s, 1, s+1, t-1) \xrightarrow{\quad k-s-t+2 \quad} \\
&(t-2, s, 1, s+1, t-2) \xrightarrow{\quad s-t+3 \quad} \cdots \xrightarrow{\quad k-s-1 \quad} (1, s, 1, s+1, 1) \xrightarrow{\quad s \quad} (0, s, 1, s+1, 1).
\end{aligned}$$

Therefore all vertices in $(0, s, 1, s+1, 1)$ are activated.

Next,

$$\begin{aligned}
(0,s,1,s+1,1) &\xrightarrow{\quad 1 \quad} (0,s,1,s,1),\\
(0,s,0,s,1) &\xrightarrow{\quad 1 \quad} (0,s,1,s,1).
\end{aligned}$$

Since both source families are active, every vertex in $(0,s,1,s,1)$ becomes activated.

Finally,

$$\begin{aligned}
&(0,s,1,s,1) \xrightarrow{\quad 2 \quad} (0,s-1,1,s,1) \xrightarrow{\quad 2 \quad} (0,s-1,1,s-1,1) \xrightarrow{\quad 3 \quad} \cdots\\
&\xrightarrow{\quad s \quad} (0,1,1,1,1) \xrightarrow{\quad s+1 \quad} (0,0,1,1,1) \xrightarrow{\quad k-s \quad} (0,0,1,1,0).
\end{aligned}$$

Thus every vertex in $(0,0,1,1,0)$, or equivalently every vertex in $X_3 \times Y_1$, becomes activated. $\square$

Combining Proposition 2.3 with Theorems 3.4 and 4.3, we obtain Theorem 1.2.

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